AC9M6A03 • Year 6 Mathematics

Algorithms, Decisions and Emerging Patterns

Create, trace, debug and generalise number-generating processes

We are learning to…

Students write complete algorithms with inputs, ordered operations, decisions, loops and stopping conditions, then analyse output sets and test generalisations.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

create and use algorithms involving a sequence of steps and decisions that use rules to generate sets of numbers; identify, interpret and explain emerging patterns

Trace an even–odd decision algorithm

1. **Input**
   n from 1 to 12
2. **Decision**
   Is n even?
3. **Even branch**
   output 3n
4. **Odd branch**
   output 3n + 1
5. **Record**
   place output in table
6. **Analyse**
   compare parity and differences

Every possible input needs a defined path. Output patterns depend on both the arithmetic rule and which inputs follow each branch.

Debug and use loops

problem	repair
no ‘no’ branch	**define action for false condition**
loop never stops	**add stopping condition**
step order ambiguous	**number instructions and define operations**
pattern from 3 outputs	test more inputs and seek reasoning
digital result only	trace sample inputs manually

A program or spreadsheet can generate many outputs, but explanation requires identifying why the pattern emerges.

Build and annotate the model

Represent algorithms, decisions and emerging patterns and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Only the yes branch defined:** Include every outcome.
- **Order of operations unspecified:** Use exact language and brackets.
- **Loop has no stopping condition:** Ensure termination.
- **Output pattern described without input rule:** Connect pattern to algorithm structure.

Quick check

- Trace inputs 1–6.
- Add a missing branch.
- Write a stopping condition.
- Describe output parity.
- Test a conjecture.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.